

Question # 1: $P(n) = 1+2+3+ \dots + n = \frac{n(n+1)}{2}$ for all integers $n \geq 1$.

Solution:

Basis step: $n=1$

$$n = \frac{n(n+1)}{2}$$

$$1 = \frac{1(1+1)}{2}$$

$$1 = \frac{1(2)}{2} = \frac{2}{2}$$

$1 = 1$ Hence $P(n)$ is true for $n=1$.

Inductive step

$n = k$

$$P(k) = 1+2+3+ \dots + k = \frac{k(k+1)}{2}$$

$n = k+1$

$$\text{Since } P(k) = 1+2+3+ \dots + k = \frac{k(k+1)}{2}$$

Add $(k+1)$ both side

$$\begin{aligned} 1+2+3+ \dots + k + (k+1) &= (k+1) + \frac{k(k+1)}{2} \\ &= (k+1) \left(1 + \frac{k}{2}\right) \\ &= (k+1) \left(\frac{2+k}{2}\right) \\ &= (k+1) \left(\frac{k+2}{2}\right) \end{aligned}$$

Hence $p(n)$ is true for $n = k + 1$.

Rough work

$$\begin{aligned} n &= \frac{n(n+1)}{2} \\ (k+1) &= (k+1) \left(\frac{k+1+1}{2}\right) \\ (k+1) &= (k+1) \left(\frac{k+2}{2}\right) \end{aligned}$$

Question #2: $1+3+5+ \dots + (2n - 1) = n^2$ for all integers $n \geq 1$.

Solution:

Basis step: $n = 1$

$$(2(1) - 1) = 1^2$$

$$(2 - 1) = 1$$

$1 = 1$ Hence $P(n)$ is true for $n=1$.

Inductive step

$n = k$

$$1+3+5+ \dots + (2k - 1) = k^2$$

$n = k+1$

$$1+3+5+ \dots + (2k - 1) = k^2$$

Add $(2k+1)$ both side

$$1+3+5+ \dots + (2k - 1) + (2k + 1) = k^2 + (2k + 1)$$

$$= k^2 + 2k + 1$$

$$= (k + 1)^2$$

Hence $p(n)$ is true for $n = k + 1$.

Rough work

$$(2(k + 1) - 1) = (k + 1)^2$$

$$2k + 2 - 1 = (k + 1)^2$$

$$2k + 1 = (k + 1)^2$$

Question # 3: $2^0 + 2^1 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all nonnegative integers.

Solution:

Basis Step $n = 0$

$$2^n = 2^{n+1} - 1$$

$$2^0 = 2^{0+1} - 1$$

$$1 = 2^1 - 1 = 2 - 1$$

$1 = 1$ Hence P(n) is true for $n=1$.

Inductive Step

$n = k$

$$2^0 + 2^1 + 2^2 + \dots + 2^k = 2^{k+1} - 1$$

$n = k+1$

Add 2^{k+1} both side

$$= 2^{k+1} + 2^{k+1} - 1$$

$$= 2^{k+1} (1 + 1) - 1$$

$$= 2^{k+1} (2) - 1$$

$$= 2^{k+1} (2^1) - 1$$

$$= 2^{k+1+1} - 1$$

$$= 2^{k+2} - 1$$

Hence p(n) is true for $n = k + 1$.

Rough work

$$2^n = 2^{n+1} - 1$$

$$2^{k+1} = 2^{k+1+1} - 1$$

$$2^{k+1} = 2^{k+2} - 1$$